Kauaʻi – Inland flooding ESF-6 playbook

Data as of 2026-10-28

County: Kauaʻi · **Hazard:** Inland flooding

Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.

1. Summary

Calculated by open code from public data · see Sources (section 8)

People in the risk area **21,404**	May need a public shelter (likely) **1,070**	May need a public shelter (low – high) **428 – 2,354**
Speak English less than well (C) **505**	Use power-dependent medical equipment (estimate) (M) **114**	People with a disability (I) **2,241**
Age 65 and older (I) **4,730**	People in households without a car (T) **919**	Hearing difficulty (C) **713**
Vision difficulty (C) **504**	Cognitive difficulty (memory, decisions) (I) **944**	Age 65 and older, living alone (S) **975**
Known 2026 shelter sites usable for this hazard **5**		

Source: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER, FEMA National Risk Index v1.20; shelter rates: section 8.

Share of people in the risk area who may need a public shelter: low 2%, likely 5%, high 11%. One person can be counted in more than one need. Page 2 explains how to read these numbers.

2. Scope: county total vs. one community

The county total is an upper bound. It counts the 25% of Census areas in the county with the highest FEMA risk score for this hazard as hit at the same time. Real events usually hit fewer communities.

Largest single community (the Census areas with the most people who may need a shelter)

Community (Census area)	People in the risk area	May need a public shelter (low – likely – high)	Speak English less than well	Use power-dependent medical equipment (estimate)	People with a disability	Age 65 and older	People in households without a car
Līhuʻe Census area 405	7,301	146 – **365** – 803	276	33	856	1,388	120
Kekaha / Waimea Census area 409	5,172	103 – **259** – 569	140	26	410	1,077	365
Wailua / Wailua Homesteads Census area 402.05	4,681	94 – **234** – 515	40	28	501	1,175	67

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

To plan for one community, use the first row.

How to read this playbook

Example row

Community (Census area)	People in the risk area	May need a public shelter (low – likely – high)
Līhuʻe Census area 405	7,301	146 – **365** – 803

Read it as: about 7,301 people live in the risk area of Līhuʻe. If this hazard hits, between 146 and 803 of them may need a public shelter; 365 is the likely number.

Census area

A neighborhood-sized area the U.S. Census uses, usually 1,200–8,000 people. The number (for example "Census area 308") is its Census ID. Each area is named after the Census place where most of its people live. "near" means no one in the area lives inside a named place; "rural near" means most people live outside named places.

Low – likely – high

The share of people in the risk area who may go to a public shelter: low 2%, likely 5%, high 11%. Where the rates come from: Team assumptions: likely 5%; low 2% (the hurricane planning factor); high 11% (the Lahaina 2023 rate).

Who has access and functional needs (CMIST)

- **C – Communication:** people who speak English less than well (they may need interpreters and translated messages), and people with hearing or vision difficulty.
- **M – Medical:** people who use power-dependent medical equipment (an estimate from HHS emPOWER).
- **I – Independence:** people with a disability, people with cognitive difficulty (memory, decisions) and people age 65 and older.
- **S – Safety and support:** people age 65 and older living alone, children and people living in group quarters (in the needs table).
- **T – Transportation:** people in households without a car.

County total vs. one community

The county total counts the 25% of Census areas in the county with the highest FEMA risk score for this hazard as hit at the same time, so it is an upper bound. Use the "Largest single community" table to plan for an event that hits one community.

Map

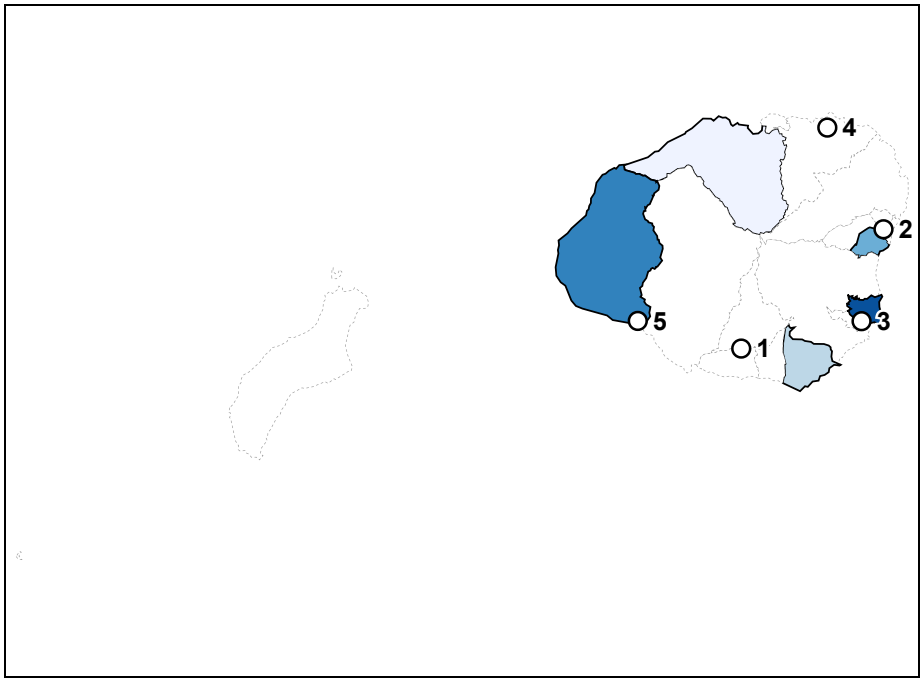
Darker blue means more people with needs, from "Lowest need" to "Highest need". A solid outline marks a Census area in the risk area; a dashed outline is outside it. Symbols: ○ usable site, △ check flood zone, ◇ flood risk not mapped, ⊠ not for this hazard. Numbers on the map match the site key.

Why population numbers can differ

People in the risk area come from ACS 2018–2022 estimates (through the CDC Social Vulnerability Index 2022). Community names use 2020 Census counts. County totals in the plan use Census estimates for 2025. Each table names its source underneath.

Every number is a planning estimate. In an event, follow county announcements.

3. Map: people with needs and known sites



Site key

1 = Kalāheo Neighborhood Center (usable)

2 = Kapa‘a Middle School (usable)

3 = Kaua‘i High School Gym (usable)

4 = Kīlauea Neighborhood Center (usable)

5 = Waimea Canyon Middle School (usable)

- Lowest need (0–310 people)
- Lower (310–607 people)
- Middle (607–636 people)
- Higher (636–940 people)
- Highest need (940–1,285 people)
- Solid outline: Census area in the risk area
- Dashed outline: outside the risk area
- Usable site
- Check flood zone
- Flood risk not mapped (FEMA zone D)
- Not for this hazard

Color = people in the risk area who speak English less than well, use power-dependent medical equipment, have a disability, or live without a car (a person can be counted more than once). No street map: Census area outlines only.

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

No 2026 activated site within a 15-minute drive (normal traffic), for sites usable for inland flooding: 6 of 17 populated Census areas. With DOE candidate schools usable for this hazard: 5 of 17. This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

Drive-time matrix, per hazard: shortest drive from the area's land point to a 2026 activated site usable for that hazard on the same island; scenario 2 adds the 5 nearest DOE candidate schools usable for it. Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

4. Known sites (opened in 2026 events)

Compiled from official county releases · verified by team

#	Status for this hazard	Site	Address	Why	Times opened in 2026
1	✓ **Usable**	Kalāheo Neighborhood Center	4480 Papalina Road, Kalāheo	Outside FEMA high-risk flood zones (zone X)	1
2	✓ **Usable**	Kapaʻa Middle School	4867 Olohena Road, Kapaʻa	Outside FEMA high-risk flood zones (zone X)	1
3	✓ **Usable**	Kauaʻi High School Gym	3577 Lala Road, Līhuʻe	Outside FEMA high-risk flood zones (zone X)	1
4	✓ **Usable**	Kīlauea Neighborhood Center	2460 Keneke Street, Kīlauea	Outside FEMA high-risk flood zones (zone X)	1
5	✓ **Usable**	Waimea Canyon Middle School	9555 Huakai Road, Waimea	Outside FEMA high-risk flood zones (zone X)	1

Source: county announcements of sites opened in 2026; FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM); tsunami evacuation zones (Hawaiʻi Statewide GIS).

Site locations: U.S. Census Geocoder, checked against Esri ArcGIS World Geocoder. Travel times: Esri ArcGIS routing, normal traffic. Powered by Esri. Geocoding and routing: Esri ArcGIS Location Services.

Candidate public schools in this county usable for this hazard: **16**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

Care facilities in hazard zones

Care facilities in hazard zones – Kauaʻi, Inland flooding (6 of 36)

Name	Type	Zone
Aloha School Early Learning Center	Childcare centre	FEMA flood zone AE
Ke Kula Niʻihau O Kekaha PCS	Childcare centre	FEMA flood zone AH
Kekaha Elementary School	Childcare centre	FEMA flood zone AH
Kekaha Head Start	Childcare centre	FEMA flood zone AE
St Theresa Early Learning Academy	Childcare centre	FEMA flood zone AE
The Koloa Early School	Childcare centre	FEMA flood zone AE

Source: CMS Provider Data Catalog (dialysis facilities, nursing homes, hospitals; facility addresses only), located with Esri ArcGIS World Geocoder; zones: State tsunami evacuation zone and FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM).

Source: Ready Keiki Locator, all pre-K sites (Office of the Lieutenant Governor, published through Hawaiʻi Statewide GIS): licensed group child care, public pre-K and charter pre-K sites only – no family child care homes; zones: State tsunami evacuation zone and FEMA flood zones (Hawaiʻi Statewide GIS, State DFIRM).

5. Needs and proposed partners

Likely estimate. Roles are **proposed – to be confirmed with partner**. The pledged column stays blank until a partner confirms it.

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Cots / sleeping space	1,071	cots	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Floor space (≤72 h)	21,404	sq ft	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Floor space (≤2 weeks)	42,808	sq ft	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Water	1,071	gal/day	American Red Cross	Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community	—	confirm with partner
Meals	2,141	meals/day	American Red Cross	Hawai‘i Department of Human Services (DHS), County Emergency Management, Hawai‘i Department of Education (DOE) – schools, The Salvation Army, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Faith-based organizations, Business community	—	confirm with partner
Toilets (≤72 h)	27	toilets	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Toilets (≤2 weeks)	54	toilets	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies	—	confirm with partner
Shelter staff	77	staff per shift	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i VOAD (Voluntary Organizations Active in Disaster), Office of Hawaiian Affairs (OHA), The Salvation Army, Philanthropy (foundations and donors), Business community	—	confirm with partner
Power outlets for power-dependent residents	6	outlets	Hawai‘i Department of Health (DOH)	County Emergency Management, Health care organizations (incl. managed health care), Utilities	—	confirm with partner
Accessible transport	919	people in no-vehicle households at risk	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Transportation (DOT), Business community, Other state and county agencies	—	confirm with partner

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Pet space – dog households	157	households with a dog	County Emergency Management	Hawai‘i VOAD (Voluntary Organizations Active in Disaster)	—	confirm with partner
Pet space – cat households	91	households with a cat	County Emergency Management	Hawai‘i VOAD (Voluntary Organizations Active in Disaster)	—	confirm with partner
Children's area	246	children in shelter	County Emergency Management	Hawai‘i Department of Human Services (DHS), Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i Department of Education (DOE) – schools, Hawai‘i State Public Library System, American Red Cross, Faith-based organizations, Other state and county agencies, Hawai‘i Department of Health (DOH), Health care organizations (incl. managed health care), Philanthropy (foundations and donors)	—	confirm with partner
Interpreters – Ilocano	10	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Interpreters – Tagalog/Filipino	6	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Spanish	2	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Japanese	2	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner

Need	How many	Unit	Proposed lead	Proposed support	Pledged by partner	Still needed
Interpreters – Marshallese	1	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner
Interpreters – Hawaiian	1	people with limited English in shelter	Hawai‘i Department of Human Services (DHS)	County Emergency Management, Hawai‘i Emergency Management Agency (HI-EMA), Hawai‘i State Public Library System, Hawai‘i Department of Health (DOH), Office of Hawaiian Affairs (OHA), Faith-based organizations, Other state and county agencies	—	confirm with partner

Source: likely estimate × planning factors (section 8; Florida Division of Emergency Management 2010; AVMA (2020 data) for pets).

Service animals stay with their handler (ADA); not counted as pets.

Also plan for (county counts, not included in the totals above)

Group	People	Detail	Date
People experiencing homelessness (Point-in-Time count)	516	sheltered 76, unsheltered 440; one-night count	2026-01-25
Visitors on island (average per day)	28,577	Kaua‘i	2025 (annual average)
People in dam evacuation zones	8,369	State dam evacuation zones	zones fetched 2026-10-28

Source: Point-in-Time counts – Bridging the Gap 2026 report (Hawai‘i, Maui, Kaua‘i; night of Jan 25, 2026) and Partners In Care 2026 count for O‘ahu as released by the City and County of Honolulu; visitors – DBEDT State of Hawai‘i Data Book 2025, Table 7.06. Details: section 8.

State dam evacuation zones show the area that could flood if a dam fails; they are planning zones, not a forecast.

Source: State of Hawai‘i dam evacuation zones (published through Hawai‘i Statewide GIS). People = share of each Census area inside a zone × its population, the same method as for the tsunami zone. Shares of people are 2020 Census block populations

whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.

Who else to plan for

- **Domestic-violence survivors:** Served through confidential providers; no counts by design.
- **Off-grid households:** No public Census-area data; county and health partners to supply.
- **People on dialysis or receiving home care:** No public Census-area data; county and health partners to supply.
- **Earthquake:** Not modeled; listed as a limitation in the plan.
- **Caregivers and personal care attendants:** Stay with the person they support; count them for cots, meals and space.
- **Behavioral health:** Mental-health staff at each shelter (ratio: section 8).
- **Niʻihau:** Privately owned island, no public shelter; plan with the landowner and Kauaʻi County.
- **Families afraid to seek shelter (including immigrant families):** The county publishes who may use shelters; trusted community messengers share it.

6. Decision points

Proposed – supports, does not replace, county plans.

When	Trigger	Action	Owner (proposed)	Support
BEFORE	Annual pre-positioning review (no official product) County (sets the date)	Mark shelters inside a FEMA Special Flood Hazard Area or zone D for site check; choose alternates.	County Emergency Management (each county)	Hawai‘i Department of Education (DOE) – schools, Hawai‘i Department of Human Services (DHS)
BEFORE	Before each season: who may use shelters (no official product) County (sets the date)	County confirms and publishes who may use shelters and what ID is needed; trusted community messengers share it.	County Emergency Management (each county)	Hawai‘i Department of Human Services (DHS), Faith-based organizations, Office of Hawaiian Affairs (OHA), Hawai‘i VOAD (Voluntary Organizations Active in Disaster)
DURING	Flash Flood Warning NOAA NWS Honolulu	Open only shelters outside the SFHA; accessible transport where roads allow; multilingual messages.	County Emergency Management (each county)	Hawai‘i Department of Transportation (DOT), American Red Cross, Hawai‘i Department of Human Services (DHS)
AFTER	Warning cancelled or all clear County / NWS / PTWC	Recovery sheltering and feeding for displaced households.	American Red Cross	Hawai‘i VOAD (Voluntary Organizations Active in Disaster), The Salvation Army, Faith-based organizations

Source: National Weather Service, Central Pacific Hurricane Center and Pacific Tsunami Warning Center product definitions (links: section 8).

7. Priority outreach and languages

Top 10 Census areas by number of people with needs

#	Community (Census area)	People in the risk area	Use power-dependent medical equipment (estimate)	People with a disability	Speak English less than well	People in households without a car	Needs total (a person can be in more than one column)
1	Līhuʻe Census area 405	7,301	33	856	276	120	1,285
2	Kekaha / Waimea Census area 409	5,172	26	410	140	365	940
3	Wailua / Wailua Homesteads Census area 402.05	4,681	28	501	40	67	636
4	Kōloa / Poʻipū Census area 406.03	2,972	19	284	49	254	607
5	Hāʻena / Hanalei Census area 401.04	1,278	7	190	0	113	310

Source: people and needs – CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022 estimates), HHS emPOWER and FEMA National Risk Index v1.20; community names – 2020 Census blocks and places, spellings from USGS GNIS.

More needs in the largest communities

Community (Census area)	Hearing difficulty	Vision difficulty	Cognitive difficulty (memory, decisions)	Age 65 and older, living alone
Līhuʻe Census area 405	297	228	326	312
Kekaha / Waimea Census area 409	81	65	135	247
Wailua / Wailua Homesteads Census area 402.05	223	127	343	226

Source: ACS 2020–2024 5-year Census-area tables B18102 (hearing), B18103 (vision), B18104 (cognitive) and B09020 (age 65 and older living alone), times the share of each area in the risk area; homes in 10+ unit buildings: CDC/ATSDR SVI 2022 (E_MUNIT, housing units).

How to reach residents

Channel	Owner (proposed)	Support (proposed)
Official alerts: Wireless Emergency Alerts / Emergency Alert System and county alert sign-ups	County Emergency Management (each county)	Hawai‘i Emergency Management Agency (HI-EMA)
Broadcast radio, including in-language programs where available	County Emergency Management (each county)	Hawai‘i Emergency Management Agency (HI-EMA)
Outdoor warning sirens: a signal to turn on radio, TV or phone alerts for instructions	Hawai‘i Emergency Management Agency (HI-EMA)	County Emergency Management
Trusted community messengers: faith-based groups, Native Hawaiian organizations, health care providers, community nonprofits	Hawai‘i Department of Human Services (DHS)	Faith-based organizations, Office of Hawaiian Affairs (OHA), Health care organizations (incl. managed health care), Hawai‘i VOAD (Voluntary Organizations Active in Disaster)
Printed notices at shelters, libraries and schools, in the county's top languages	County Emergency Management (each county)	Hawai‘i State Public Library System, Hawai‘i Department of Education (DOE) – schools

Source: proposed channels and formats in the plan's communication pathways – supports, does not replace, county and State public-information plans.

Populated Census areas whose land point is farther than 3,400 ft from every outdoor siren (HI-EMA manufacturer rating; actual range varies): **14**.

Populated Census areas with less than 50% of people within rated siren range (4): prioritise WEA, radio and trusted messengers. Kapa‘a (403.01); Lāwa‘i / ‘Ōma‘o (406.04); Princeville (401.06); Wailua Homesteads / Kapa‘a (402.04)

Source: Hawai‘i Emergency Management Agency public siren map (locations of permanent outdoor warning sirens; no live status). Rated range: 3,400 feet, the manufacturer rating published by the Hawai‘i Emergency Management Agency; the real range varies with terrain, buildings and weather. In range: within 3,400 feet of a siren, the same method as for the tsunami zone. Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.

Accessible formats for every message:

- **ASL video:** short American Sign Language video of each message
- **Captions:** captions on every video and broadcast
- **Large print:** printed notices in large print
- **Plain language:** short sentences, one action at a time
- **Screen-reader-friendly text:** plain text or tagged web pages, not images of text

Languages: people age 5 and older who speak English less than "very well" (Kauaʻi + Maui)

Kauaʻi and Maui share one language list: the Census survey reports both counties (and Kalawao) as one area, so they cannot be separated.

Language	People	Margin of error (90%)
Ilocano	9,131	±1,439
Tagalog/Filipino	5,074	±926
Spanish	1,711	±554
Japanese	1,339	±417
Marshallese	633	±488
Hawaiian	515	±289

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawaiʻi Data Book 2024, Table 1.44).

Draft messages are available statewide in Chinese, Hawaiian, Ilocano, Japanese, Korean, Pidgin, Spanish, Tagalog and Vietnamese; Chuukese and Marshallese need translators.

All languages

All languages in the survey for Kauaʻi + Maui: people age 5 and older who speak English less than "very well"

Language	People	Margin of error (90%)	Reliability
Ilocano	9,131	±1,439	
Tagalog/Filipino	5,074	±926	
Spanish	1,711	±554	
Japanese	1,339	±417	
Marshallese	633	±488	low reliability
Hawaiian	515	±289	
Chinese	510	±266	
Korean	360	±210	
Vietnamese	359	±216	
Tongan	320	±212	low reliability
Other Eastern Malayo-Polynesian languages	277	±178	
Chuukese	too few to estimate		low reliability
Cebuano and other Philippine languages	259	±192	low reliability
Portuguese	too few to estimate		low reliability
Pidgin (Hawaiʻi Creole English)	220	±192	low reliability
German	too few to estimate		low reliability
Thai	too few to estimate		low reliability
Chamorro	too few to estimate		low reliability
Sinhala	too few to estimate		low reliability
Greek	too few to estimate		low reliability
Russian	too few to estimate		low reliability
Dutch	too few to estimate		low reliability
Khmer	too few to estimate		low reliability

Language	People	Margin of error (90%)	Reliability
Samoan	too few to estimate		low reliability
French	too few to estimate		low reliability
Hungarian	too few to estimate		low reliability
Italian	too few to estimate		low reliability
Latvian	too few to estimate		low reliability
Afrikaans	too few to estimate		low reliability
Akan (including Twi)	too few to estimate		low reliability
Arabic	too few to estimate		low reliability
Bulgarian	too few to estimate		low reliability
Danish	too few to estimate		low reliability
Farsi	too few to estimate		low reliability
Finnish	too few to estimate		low reliability
Hebrew	too few to estimate		low reliability
Hindi	too few to estimate		low reliability
Hmong	too few to estimate		low reliability
Indonesian	too few to estimate		low reliability
Lao	too few to estimate		low reliability
Malay	too few to estimate		low reliability
Mongolian	too few to estimate		low reliability
Norwegian	too few to estimate		low reliability
Other Indo-European languages	too few to estimate		low reliability
Other and unspecified languages	too few to estimate		low reliability
Polish	too few to estimate		low reliability
Punjabi	too few to estimate		low reliability
Serbian	too few to estimate		low reliability
Swedish	too few to estimate		low reliability
Swiss German	too few to estimate		low reliability
Tamil	too few to estimate		low reliability
Turkish	too few to estimate		low reliability
Yiddish	too few to estimate		low reliability

Source: ACS PUMS 2019–2023 5-year (validated against the DBEDT State of Hawai‘i Data Book 2024, Table 1.44).

Low reliability: margin of error large compared with the estimate (coefficient of variation above 40%), or too few to estimate (fewer than 100 people or 10 survey responses).

Pre-approved message templates: none yet; no machine translation without human review.

8. Sources and assumptions

Planning factors

Factor	Value	Source or team assumption
Shelter-seeking rate – Hurricane	2% / 3% / 5%	South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%)
Shelter-seeking rate – Wildfire	2% / 5% / 11.2%	high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are Team assumption.
Shelter-seeking rate – Tsunami	2% / 5% / 11%	Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Inland flooding	2% / 5% / 11%	Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Shelter-seeking rate – Coastal flooding	2% / 5% / 11%	Team assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate
Footprint – Hurricane	Every Census area	Hurricanes affect whole islands (Lala, Lowell 2026).
Footprint – Tsunami	Share living inside the tsunami evacuation zone	Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately.
Footprint – Wildfire	Top 25% of Census areas by FEMA risk score	Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption).
Footprint – Inland flooding	Top 25% of Census areas by FEMA risk score	Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption).
Footprint – Coastal flooding	Top 25% of Census areas by FEMA risk score	Counties lead mass care, so each county plans for its own highest-risk tracts (Team assumption).
Floor sqft 72h	20	Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h
Floor sqft 2wk	40	Florida DEM (2010): short-term shelter, <=2 weeks
Water gal per day	1	Florida DEM (2010)
Meals per day	2	Florida DEM (2010): two hot meals a day plus snacks

Factor	Value	Source or team assumption
Persons per toilet short term	20	Florida DEM (2010): short-term shelter, <=2 weeks
Power outlets per power dependent person	1	Team assumption
Persons per toilet 72h	40	Florida DEM (2010): emergency evacuation shelter, <=72 h
Staffing ratios (one staff member per N residents)	client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12	One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations.
Dog owning household share	0.446	AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27.
Cat owning household share	0.26	AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'.

Data sources

Source	Fetched	URL / file
CDC/ATSDR Social Vulnerability Index 2022, census tracts	2026-10-27	https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2
FEMA National Risk Index v1.20, census tracts – Hurricane	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_Rating/FeatureServer/0
FEMA National Risk Index v1.20, census tracts – Wildfire	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Rating/FeatureServer/0
FEMA National Risk Index v1.20, census tracts – Tsunami	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_Rating/FeatureServer/0
FEMA National Risk Index v1.20, census tracts – Inland Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_Index_Rating/FeatureServer/0

Source	Fetched	URL / file
FEMA National Risk Index v1.20, census tracts – Coastal Flooding	2026-10-27	https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_Index_Rating/FeatureServer/0
Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2
Extreme Tsunami Evacuation Zones (Statewide GIS)	2026-10-27	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12
HHS emPOWER – electricity-dependent DME, county level	2026-10-27	https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2
ACS 2020-2024 5-year, tract tables B18102 (hearing), B18103 (vision), B18104 (cognitive), B09020 (age 65+ living alone)	2026-10-26	https://data.census.gov/api/access/data/table?g=040XX00US15%241400000&id=ACSDT5Y2024.B18102
Share of people in a zone (tsunami, dam evacuation zones, siren range)		Shares of people are 2020 Census block populations whose block internal point is inside (Census TIGER internal points), divided by the Census area's 2020 block population; the share is then applied to the area's population.
Hawai‘i Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26)	2026-10-28	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6
County shelter activations 2026 (official county releases)	2026-10-27	County shelter activation table: one official county source per row
2020 Census Block Assignment File, place/CDP (tract community labels)	2026-10-27	https://www2.census.gov/geo/docs/maps-data/data/baf2020/BlockAssign_ST15_HI.zip
2020 Census PL 94-171 block population P1_001N (tract community labels)	2026-10-27	https://www2.census.gov/programs-surveys/decennial/2020/data/01-Redistricting_File--PL_94-171/Hawaii/hi2020.pl.zip
ACS PUMS 2019-2023 5-year (detailed languages)	2026-10-27	https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip
DBEDT State of Hawai‘i Data Book 2024, Table 1.44 (validation)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf
Partner list: Hawai‘i DHS – ESF-6 challenge statement (2026)	2026-10-15	Hawai‘i DHS – ESF-6 challenge statement (2026)
pre season hurricane		https://www.nhc.noaa.gov/cphcmission.php
hurricane watch		https://www.nhc.noaa.gov/aboutgloss.shtml
hurricane warning		https://www.nhc.noaa.gov/aboutgloss.shtml
tsunami warning		https://www.tsunami.gov/?page=message_definitions
flash flood warning		https://www.weather.gov/safety/flood-watch-warning
red flag warning		https://www.weather.gov/safety/wildfire-ww

Source	Fetched	URL / file
coastal flood warning		https://www.weather.gov/okx/marinedef
coastal flood warning in effect		https://www.weather.gov/okx/marinedef
Ready Keiki Locator – All Pre-K Sites (Office of the Lieutenant Governor, hosted by Hawai‘i Statewide GIS)	2026-10-28	https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/Ready_Keiki_GIS/FeatureServer/0
Hawai‘i Emergency Management Agency – all-hazard outdoor warning sirens (public siren layer; locations only)	2026-10-28	https://services9.arcgis.com/aKxrz4vDVjfUwBWJ/arcgis/rest/services/public_siren_status_layer_view/FeatureServer/0
State of Hawai‘i dam evacuation zones (hosted by Hawai‘i Statewide GIS)	2026-10-28	https://services.arcgis.com/HQ0xoN0EzDPBOEci/arcgis/rest/services/DamEvacZones_joined/FeatureServer/0
Hawai‘i Emergency Management Agency – siren audible radius (3,400 feet, manufacturer rating)	2026-10-28	https://dod.hawaii.gov/hiema/all-hazard-statewide-outdoor-warning-siren-system/
Tsunami Safe Zones (Hawai‘i Statewide GIS, Hazards layer 13) (Hawai‘i Island March 2025 (corrected November 2025), O‘ahu April 2025, Kaua‘i March 2025, Maui, Moloka‘i and Lāna‘i 2013)	2026-10-30	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/13
Fire Risk Areas (DLNR Division of Forestry and Wildlife, via Hawai‘i Statewide GIS, Hazards layer 7) (2007; boundaries confirmed by DLNR-DOFAW 2017 and 2022)	2026-10-30	https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/7
Sea Level Rise – Potential Flooded Highways, 3.2 ft scenario (Hawai‘i Sea Level Rise Viewer, PacIOOS, via Hawai‘i Statewide GIS, Climate layer 71) (Hawai‘i Sea Level Rise Vulnerability and Adaptation Report, 2017)	2026-10-30	https://geodata.hawaii.gov/arcgis/rest/services/Climate/MapServer/71

Source	Fetched	URL / file
Pacific Disaster Center – Hawaii Emergency Shelters (capacity: American Red Cross Hawai‘i State Chapter) (ground condition 2009-10-08 (PDC metadata))	2026-10-30	https://services1.arcgis.com/x4h61KaW16vFs7PM/arcgis/rest/services/Potential_Emergency_Shelters_PDC/FeatureServer/36
Hawai‘i DOT Highway Performance Monitoring System road centrelines (Hawai‘i Statewide GIS, Transportation layer 12)	2026-10-30	https://geodata.hawaii.gov/arcgis/rest/services/Transportation/MapServer/12
Hawai‘i DOT surveyed mile markers (road-closure point, Route 11 in Ka‘ū)	2026-10-30	https://geodata.hawaii.gov/arcgis/rest/services/Transportation/MapServer/14
DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census)	2026-10-27	https://files.hawaii.gov/dbedt/economic/databook/db2025/DB2025_section07.xlsx
CMS Provider Data Catalog – Dialysis Facility – Listing by Facility		https://data.cms.gov/provider-data/dataset/23ew-n7w9
CMS Provider Data Catalog – Nursing Homes – Provider Information		https://data.cms.gov/provider-data/dataset/4pq5-n9py
CMS Provider Data Catalog – Hospital General Information		https://data.cms.gov/provider-data/dataset/xubh-q36u
Esri ArcGIS World Geocoder (site check, care facilities) and ArcGIS routing (drive and walk times, normal traffic). Powered by Esri.		https://developers.arcgis.com/rest/

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.

9. Results

- **Coverage:** 436 of 436 Census areas have needs data (FEMA risk data joined for 435 of 436). Source: coverage check of the needs layer (CDC SVI 2022, FEMA NRI v1.20).
- **Backtest – Lahaina wildfire (2023-08-11):** observed 1,418 people in shelters; model 254 / 635 / 1,423 (low / likely / high). This is a calibration event, not validation: the high rate was set from it. Source: Lahaina 2023 backtest (Maui Now count of people in shelters).

10. Stress tests – what could break

Stress test, not a forecast. Each test changes one thing in the plan (a site closes, two hazards strike together, a road is cut, more people come) and counts who is affected, from the plan's own data. Deterministic code computes every number from the plan's data; people review the results.

Top 10 for Kaua‘i – Inland flooding and all-hazard tests, ranked by people affected (at most two from one test). The full list, with every test, is in the app (Stress tests).

1. **Stranded visitors for 3 or 7 days – Kaua‘i** (test 21: Stranded visitors)
   28,577 visitors on an average day (island level).
   By length of stay: 3 days: 171,462 meals, 85,731 gallons of water; 7 days: 400,078 meals, 200,039 gallons of water · Visitors on an average day (2025): 28,577
   Sources: DBEDT State of Hawai‘i Data Book 2025, Table 7.06 (average daily visitor census, 2025); Florida Division of Emergency Management, Sheltering Guidance Aid (2010). Team assumptions: Every visitor on the island is stranded and needs meals and water from mass care (upper bound): 100%.
2. **All four counties at once (inland flooding)** (test 9: Several counties at once)
   25,452 people who may need a public shelter (likely), counties together (cots: 25,454; shelter staff per shift: 1,723).
   Cots: 25,454 · Counties: Kauai, Honolulu, Maui, Hawaii · People with limited English in shelter, by language (interpreters): Chinese: 100; Chuukese: 23; Hawaiian: 11; Ilocano: 152; Japanese: 82; Korean: 40; Marshallese: 11; Pidgin (Hawai‘i Creole English): 11; Spanish: 13; Tagalog/Filipino: 123; Vietnamese: 26 · Meals per day: 50,906 · People in shelter (likely): 25,452
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).
3. **Kaua‘i and Honolulu at once (inland flooding)** (test 9: Several counties at once)
   19,543 people who may need a public shelter (likely), counties together (cots: 19,545; shelter staff per shift: 1,322).
   Cots: 19,545 · Counties: Kauai, Honolulu · People with limited English in shelter, by language (interpreters): Chinese: 96; Chuukese: 23; Hawaiian: 7; Ilocano: 111; Japanese: 71; Korean: 40; Marshallese: 6; Spanish: 2; Tagalog/Filipino: 106; Vietnamese: 26 · Meals per day: 39,088 · People in shelter (likely): 19,543
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; Florida Division of Emergency Management, Sheltering Guidance Aid (2010); ACS PUMS 2019–2023 5-year (limited-English speakers by language); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions).

4. **Lose Kapa‘a Middle School and Kaua‘i High School Gym (inland flooding)** (test 2: Two sites lost)
   11,982 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 2; people more than losing either site alone: 4,681).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 2 · Drive limit (minutes): 15 · Island: Kaua‘i · People more than losing either site alone: 4,681
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
5. **All 'flooded highway' segments (sea level rise 3.2 ft) closed – Kaua‘i** (test 24: Road closures (driving))
   9,151 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 0; most minutes added to any known site: 21.7).
   Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 0 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 4
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS); Hawai‘i Sea Level Rise Viewer, potential flooded highways at 3.2 ft (PacIOOS, via Hawai‘i Statewide GIS). Team assumptions: Every flooded-highway segment on the island closed at once (sea level rise 3.2 ft scenario): all segments.
6. **Kūhiō Highway (Route 560) near the Hanalei bridge** (test 24: Road closures (driving))
   7,793 people in areas with a longer drive, fewer known sites within 45 minutes, or cut off (Census areas cut off from every 2026 site: 1; most minutes added to any known site: 0).
   Census areas cut off (over the limit or no route): 0 · Census areas cut off from every 2026 site: 1 · Census areas losing known sites within reach: 3 · Census areas with no route at all: 0 · Census areas routed: 17
   Sources: Esri ArcGIS routing with road closures (build time); highway centrelines: Hawai‘i DOT HPMS and surveyed mile markers (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time); county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Hawai‘i DOE public schools (Hawai‘i Statewide GIS). Team assumptions: Closure location along the highway (a stretch of about 0.8 km): team assumption.
7. **Lose Kaua‘i High School Gym (inland flooding)** (test 1: One site lost)
   7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kaua‘i · May need a public shelter (likely): 365.1 · Site: Kaua‘i High School Gym
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawai‘i Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).

8. **Lose Kalāheo Neighborhood Center and Kauaʻi High School Gym (inland flooding)** (test 2: Two sites lost)
   7,301 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1; people more than losing either site alone: 0).
   Census areas left with no usable 2026 site on the island: 0 · Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kauaʻi · People more than losing either site alone: 0
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
9. **Lose Kapaʻa Middle School (inland flooding)** (test 1: One site lost)
   4,681 people in the risk area pushed outside a 15-minute drive to a usable 2026 site (Census areas pushed outside a 15-minute drive: 1).
   Census areas pushed outside a 15-minute drive: 1 · Drive limit (minutes): 15 · Island: Kauaʻi · May need a public shelter (likely): 234.1 · Site: Kapaʻa Middle School
   Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); Esri ArcGIS routing (drive-time matrix, normal traffic, build time).
10. **Hurricane and inland flooding together – Kauaʻi** (test 5: Hurricane and inland flooding together)
    2,633 people who may need a public shelter (likely) (2026 sites usable for every hazard in the pair: 5; site gap: 6).
    Census areas: 17 · People if each hazard is counted separately: 94,915 · People in either footprint (counted once): 73,511 · May need a public shelter (likely): 2,633 · Site gap: 6
    Sources: CDC/ATSDR Social Vulnerability Index 2022 (ACS 2018–2022), HHS emPOWER, FEMA National Risk Index; 2020 Census blocks; county announcements of sites opened in 2026; FEMA flood zones and tsunami evacuation zones (Hawaiʻi Statewide GIS); shelter-seeking rates in the plan (South Carolina EMD 2026 Annex G; Maui 2023 observed; team assumptions). Team assumptions: Sites needed at a given number of people per site (capacity unknown): 250; A person in two footprints is counted once, at the higher rate: highest rate.

This is not a statement that the area has no shelter: counties designate and open shelters for each event, and many public buildings are not on the 2026 list.

11. Questions for counties

These need county knowledge. The plan has no data for them, so it shows no numbers.

1. **Shelter buildings in an earthquake or a strong hurricane:** Which shelter buildings have been assessed for earthquake and hurricane-wind resilience, and which need a check before they are used after a strong event?
2. **Resupply for Molokaʻi and Lānaʻi:** How would water, food, cots, fuel and medical supplies reach Molokaʻi and Lānaʻi if the harbours or airports close for several days, and who decides the order of shipments?

3. **Cell network outages:** Which areas lose mobile phone service first in a long power outage or after wind damage, and how will warnings and shelter information reach them (radio, sirens, door-to-door)?
4. **Schools used as shelters:** When a school is used as a shelter, how long can it stay open before classes must restart, and where do residents go next?
5. **Partner capacity:** Could each county collect partner capacity (cots, meals, staff, vehicles, interpreters) with a county-run form such as Survey123, with no personal data, so the plan can show pledged capacity instead of 'confirm with partner'?